

Lec-15 $\Rightarrow$ algorithms $\Rightarrow$ grade 2025
course

0-14

Lec_15

Practice Questions

Comparison of fn.

compare the functions:-

$$(\log n)^{\log n} > n$$

1.) n vs $\log n^{\log n}$

n and $\log n^{\log n}$

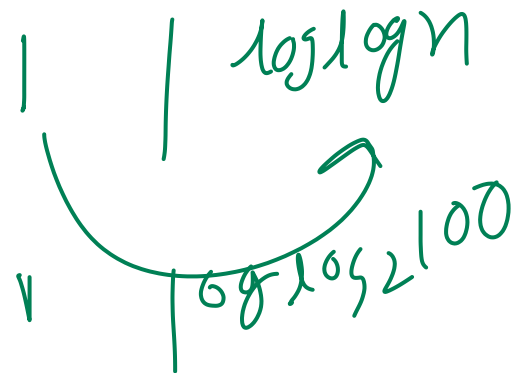
n

$(\log n)^{\log n}$

Taking $\log$ both side.
 $\log_2 n$ | $\log_2 (\log n)^{\log n}$

~~$\log n$~~ | ~~$\log n \times \log \log n$~~

$$n = \frac{100}{2}$$



~~$\log n$~~
 ~~$\log \log n$~~

2.) $n^n, 2^n, n!$

$$n^2 \quad \boxed{2^n > n^2}$$

$$n^2 \quad | \quad 2^n$$

$$\log_2 n^2 \quad | \quad \log_2 2^n$$

$$2 \log_2 n \quad | \quad \underline{n \log_2 2}$$

$$\underline{2 \log_2 n} \quad | \quad \underline{n}$$

$$\underline{2^n}$$

$$2^n \quad | \quad \underline{n!}$$

$$2^n = 2 \cdot 2 \cdot 2 \cdot 2 \cdot \dots \quad n \text{ times}$$

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 1$$

$$\text{Ex: } n = \underline{100}$$

$$\boxed{n! > 2^n}$$

$$2^n, \quad \underline{n^2}, \quad \underline{n!}$$

$$n! > 2^n$$

$$2^n > n^2$$

Ans

$$\boxed{n! > 2^n > n^2}$$

3.) $\log_2 n, \log_4 n$

$$\log_2 n \quad | \quad \log_4 n$$

$$\log_2 n \quad | \quad \frac{\log_2 n}{\log_2 4}$$

$$\log_2 n \quad | \quad \frac{\log_2 n}{2}$$

$$\log_2 n \quad | \quad \log_2 n \Rightarrow \textcircled{2}$$

Base changing Formula
In Log :-

$$\log_a b = \frac{\log_c b}{\log_c a}$$

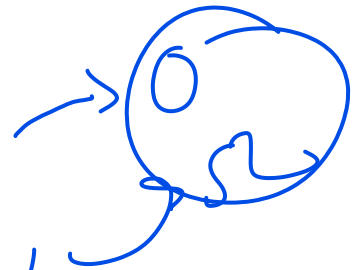
value $\Rightarrow \log_2 > \log_4 n$

Asymptotically $\Rightarrow$ Both are equal
Unit

$\Downarrow$
equal is there

$$\log_2 n = \Theta(\log_4 n)$$

$$\log_4 n = \Theta(\log_2 n)$$



4.) $2^n, 3^n$

$$2^n = \underbrace{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot \dots}_{n \text{ times}}$$

$$3^n = \underbrace{3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot \dots}_{n \text{ times}}$$

$$3^n > 2^n$$

$$2^n \times 3^n > 2^n$$

$$\cancel{2^n} \cdot \overline{2^n} (1.5)^n > \cancel{2^n}$$

$$\underline{(1.5)^n > 1}$$

$$\boxed{3^n > 2^n}$$

5.) State true or false

T/F

①

$100000 = \theta(1)$

$\Theta \Rightarrow$

$f(n) \leq c_1 g(n)$

and

$f(n) \geq c_2 g(n)$

True



$10^5 \leq c_1(1)$

$\Rightarrow c_1 = 10^6$

$10^5 \geq c_2(1) \Rightarrow c_2 = 1$

we can choose

②

$$f(n) \leq \underline{c \cdot g(n)}$$

$$n = 1$$

$$n = 2,$$

$$n = 4$$

$$f = 10$$

$$f(x)$$

$\frac{1}{2}$

1/2

rd



$$\frac{1}{n} \leq$$

C(1)

d) if $n < 4$
 then $n^n = O(n^4)$

True

True

given,

$x < y$

$$n = o(y)$$

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$$x = o(y)$$

$$n^x < n^y$$

$$T_n = O(n^4)$$



$$n^{\frac{1}{2}} = O(n^{\frac{1}{4}})$$

$$\frac{1}{n} = \theta(1)$$

T/F

4.)
false

$$f(n) \leq c_1 g(n) \text{ and } f(n) \geq c_2 g(n)$$

$$\frac{1}{n} \leq c_1(1)$$

$$c_1 = 2$$

$$\frac{1}{n} \geq c_2(1), c_2 =$$

$$\frac{1}{n} \leq 1$$

5.)

$$2^{n+1} = \theta(2^n)$$

$$f(n) \leq c_1 g(n) \text{ and } f(n) \geq c_2 g(n)$$

$$f(n) \geq c_2 g(n)$$

True

$$2^{n+1} \geq 2^n$$

$$2^{n+1} \leq c_1 2^n$$

$$2^n \cdot 2 \leq c_1 2^n$$

$$2^n \cdot 2 \leq 4 \cdot 2^n$$

$$c_1 = 4$$

(C) ~~f(n)~~ $f(n)$ is a given positive function
state T/F

False

In
case of
decreasing
function

$$\leq f(n) \leq O([f(n)]^2)$$

$$f(n) \leq C_0 [f(n)]^2$$

$$\frac{1}{n^2} \leq C \left[\frac{1}{n^2} \right]^2$$

$$\frac{1}{n^2} \leq C \frac{1}{n^4}$$

$$1 \leq C \times \frac{1}{n^2}$$

$\Rightarrow$ complexity claim

$$f(n) = \frac{1}{n^2}$$

$n=10$

$\frac{1}{n^2}$

$\frac{1}{10^2}$

$\frac{1}{100}$

7.

is

$$f(n) = O(g(n))$$



$$f(n) \leq C \cdot g(n)$$



$$2^{f(n)} \leq C \cdot g(n)$$

$$f(n) = n \log n$$

$$2^{f(n)} = O(2^{g(n)})$$

$$f(n) = n, \quad g(n) = n^2$$

$$n \leq n^2$$

$$2^n \leq n^2$$

$$f(n) \leq C \cdot g(n)$$

$$2^n \leq C \cdot n \Rightarrow C = 4$$

True

$$f(n) = 2^n$$

$$g(n) = n$$

$$2^{f(n)} = 2^{2^n}$$

$$g(n) = \frac{2^n}{2}$$

$$2^{f(n)} \leq C \cdot 2^{g(n)}$$

$$2^{2^n} \leq C \cdot 2^n$$

$$2^{n+n} \leq C \cdot 2^n$$

$$2^{2n} \leq C \cdot 2^n$$

$$2^n \leq C$$

$$\boxed{2^n \in C} \times$$

Note!

False

if $f(n) = O(g(n))$.

then

$$2^{f(n)} = O(2^{g(n)})$$

!!

$\Rightarrow$ not necessary

Thanks

